

Mô hình hóa Thông tin

Analysis: Information Modeling

Trí tuệ Nhân tạo cho Kế toán (AI in Accounting)

Khoa Kế toán - Kiểm toán

Bài giảng Thực hành 06

Góc nhìn Chuyên gia (Professional Insight)

- **Mô hình hóa Thông tin:** Không chỉ đơn giản là việc liên kết các bảng dữ liệu, mà là nghệ thuật cấu trúc hóa dữ liệu để trả lời chính xác các câu hỏi kinh doanh.
- **Khả năng mở rộng (Scalability):** Một mô hình dữ liệu tốt (ví dụ: Star Schema) cho phép báo cáo chạy nhanh hơn, dễ bảo trì hơn và hạn chế rủi ro sai lệch dữ liệu.
- **Tư duy Hệ thống:** Kế toán viên hiện đại cần tư duy giống như một Data Architect để xây dựng nền tảng vững chắc cho mọi phân tích phía sau.

Lộ trình Chương (Chapter Roadmap)

- **LO 6.1:** Khái niệm Nền tảng của Mô hình hóa Thông tin.
- **LO 6.2:** Áp dụng Các Thuật toán Mô hình hóa Thông tin (7 Mẫu cơ bản).
- **LO 6.3:** Sáu Mẫu Mô hình hóa cho Cấu trúc Dữ liệu Kế toán (Star Schema).

6.1 Mô hình hóa Thông tin là gì?

- **Định nghĩa:** Là quá trình cấu trúc và sắp xếp lại dữ liệu thô thành thông tin có ý nghĩa thông qua các thuật toán và mối quan hệ giữa các bảng.
- **Quy trình Mô hình hóa (The Information Modeling Process):** Chuyển từ Dữ liệu (Data) → Thuật toán (Algorithms) → Thông tin (Information).
- **Mục tiêu:** Đảm bảo tính toàn vẹn, tính nhất quán và dễ dàng truy xuất thông tin phục vụ cho Bảng điều khiển (Dashboards) và Báo cáo.

ILLUSTRATION 6.1

Information Modeling

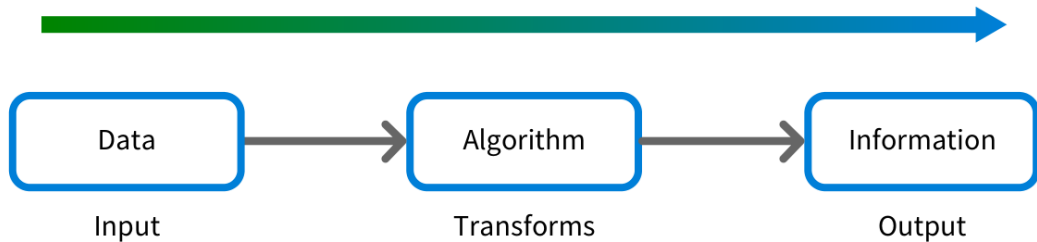


ILLUSTRATION 6.2

IF the NetAmount of a transaction is smaller than \$1,000,
THEN the TransactionSize is S (small),
ELSE
IF the NetAmount of a transaction is at least \$1,000 but smaller than \$2,500,
THEN the TransactionSize is M (medium),
ELSE the transaction size is L (large).

Algorithm for TransactionSize

Amount - Discount

Algorithm for NetAmount

ID	Date	Product	Amount	Discount	Customer	SalesPerson	NetAmount	TransactionSize
1	1/3/2022	Dishwasher	\$ 825	0	Jeremy	Ed	\$ 825	S
2	1/3/2022	Refrigerator	\$ 2,500	100	Jane	Juana	\$ 2,400	M
3	1/3/2022	Refrigerator	\$ 2,500	250	Molly	Hakeem	\$ 2,250	M
4	1/4/2022	Cooktop	\$ 1,250	75	Whitney	Ed	\$ 1,175	M
5	1/4/2022	Dishwasher	\$ 900	125	Zuzu	Juana	\$ 775	S
6	1/4/2022	Cooktop	\$ 1,500	250	Carmen	Cindy	\$ 1,250	M
7	1/4/2022	Oven	\$ 1,500	150	Ben	Hakeem	\$ 1,350	M
8	1/4/2022	Cooktop	\$ 1,000	50	Lebron	Juana	\$ 950	S
9	1/4/2022	Cooktop	\$ 900	0	Trisha	Hakeem	\$ 900	S
10	1/5/2022	Dishwasher	\$ 500	50	Xuomin	Cindy	\$ 450	S

Data

Information

ILLUSTRATION 6.3

Measure	Algorithm
TotalNetAmount	Sum of all the values in the NetAmount column.
NumberOfLargeTransactions	Count the number of cells in the TransactionSize column that have the value “L.”

ILLUSTRATION 6.4

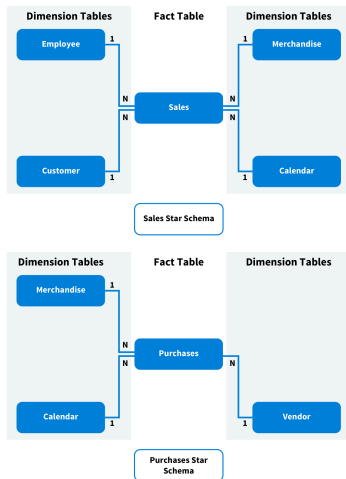


ILLUSTRATION 6.5

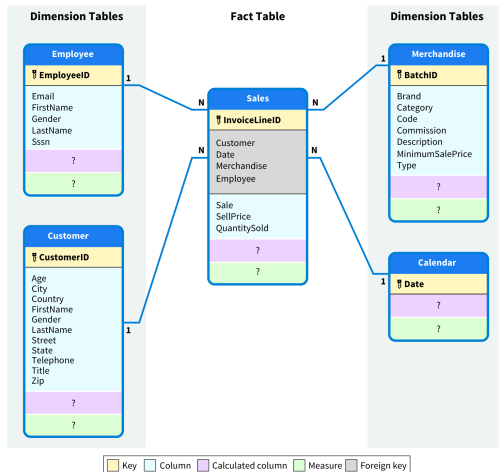


ILLUSTRATION 6.6

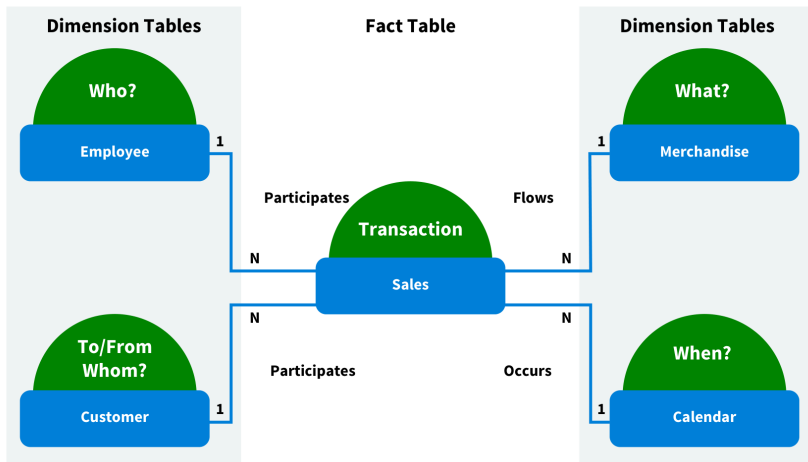


ILLUSTRATION 6.7

Table: Sales

(A)	Calculated Column	Algorithm
	Revenue	$SellPrice \times QuantitySold$

(B)

InvoiceLineID	Sale	Date	Merchandise	Customer	Employee	SellPrice	QuantitySold	Revenue
1	1	1/5/23	8	223	7	60	12	720
2	1	1/5/23	9	223	7	20	4	80
3	1	1/5/23	16	223	7	320	1	320
4	2	1/6/23	19	80968	635	9	40	360
5	2	1/6/23	20	80968	635	20	40	800

Result of Algorithm

$SellPrice \times QuantitySold$

ILLUSTRATION 6.8

Table: Customer

(A)	Calculated Column	Algorithm
	Address	Street & “,” & City & “,” & State & “,” & Zip & “,” & Country

Street + City + State + Zip + Country

Result of Algorithm

(B)

CustomerID	Street	City	State	Zip	Country	Address
1	3462 MacLaren Street	Ottawa	ON	K1P 5M7	CA	3462 MacLaren Street,Ottawa,ON,K1P 5M7,CA
2	2608 Laurel Lane	Terminal	TX	79703	US	2608 Laurel Lane,Terminal,TX,79703,US
3	4679 Byrd Lane	Albuquerque	NM	87102	US	4679 Byrd Lane,Albuquerque,NM,87102,US
4	975 Ambrester Drive	Malibu	CA	90265	US	975 Ambrester Drive,Malibu,CA,90265,US
5	3251 St. John Street	Bruno	SK	S4P 3Y2	CA	3251 St. John Street,Bruno,SK,S4P 3Y2,CA

ILLUSTRATION 6.9

Table: Customer

Calculated Column

Name

Algorithm

LastName & “,” & FirstName

Table: Vendor

Calculated Column

Name

Algorithm

Lastname & “,” & Firstname

ILLUSTRATION 6.10

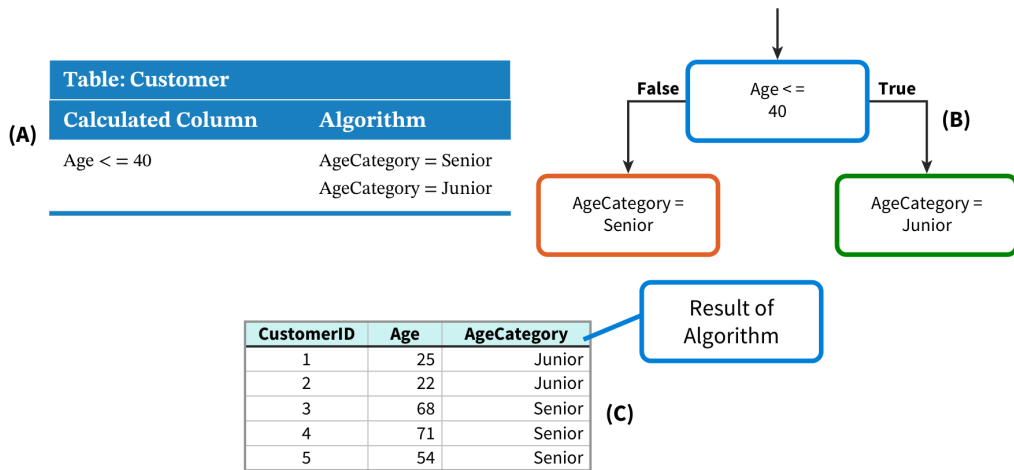
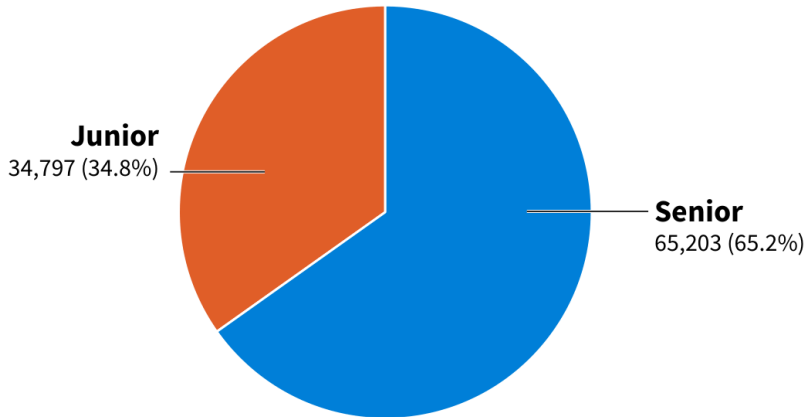


ILLUSTRATION 6.11

Relative Importance of Age Categories



Thuật toán và Phương pháp tiếp cận

- **Thuật toán (Algorithms):** Là tập hợp các quy tắc tính toán (ví dụ: Tính biên lợi nhuận = Doanh thu - Giá vốn).
- **Phương pháp tiếp cận có cấu trúc (Structured Approach):**
 - Hiểu rõ yêu cầu đầu ra (Output).
 - Xác định dữ liệu đầu vào (Input).
 - Xây dựng các bước xử lý logic (Processing).

ILLUSTRATION 6.12

Table: Sales

Calculated Column

Algorithm

COGS

$\text{Sales.QuantitySold} \times \text{Purchases.BuyPrice}$

ILLUSTRATION 6.13

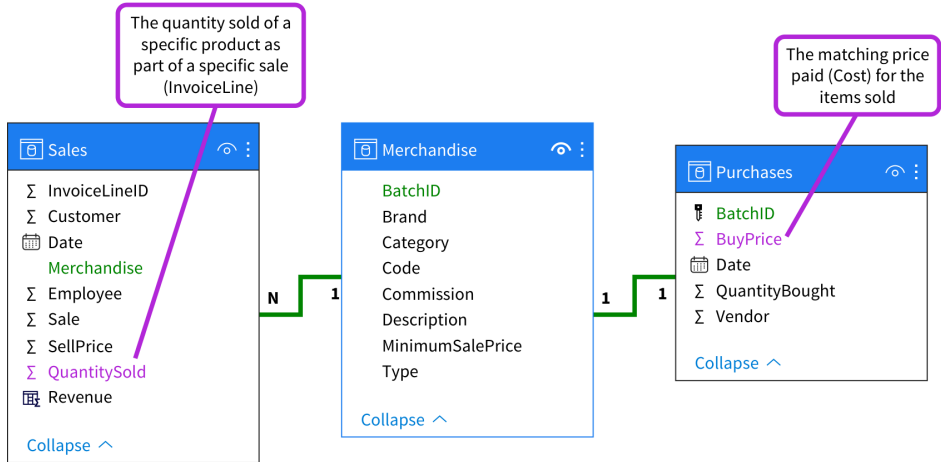


ILLUSTRATION 6.14

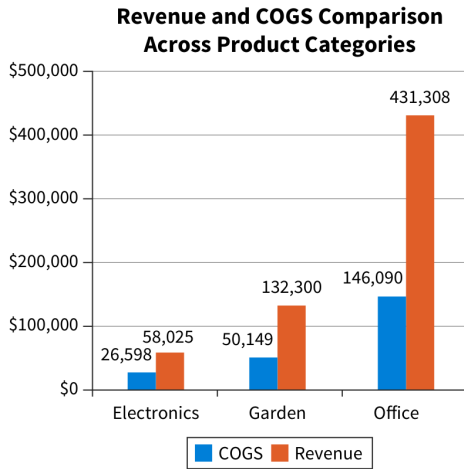


ILLUSTRATION 6.15

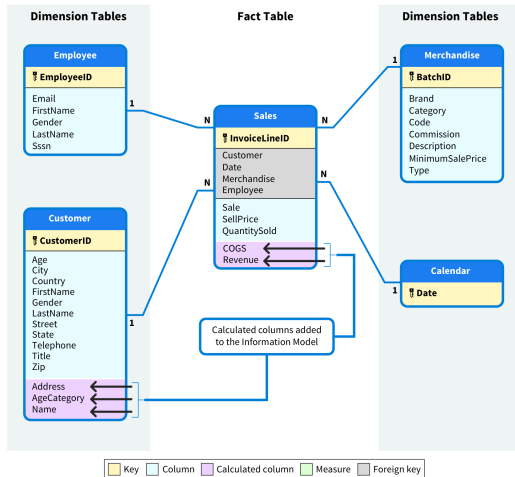


ILLUSTRATION 6.16

Table: Sales

Measure

Algorithm

TotalQuantitySold

SUM(QuantitySold)

Table: Sales

Measure

Algorithm

TotalRevenue

SUM(Revenue)

Table: Customer

Measure

Algorithm

NumberOfCustomers

COUNT(CustomerID)

ILLUSTRATION 6.17

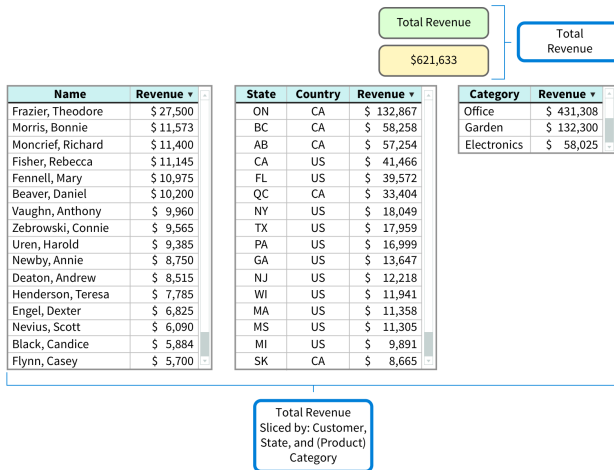


ILLUSTRATION 6.18

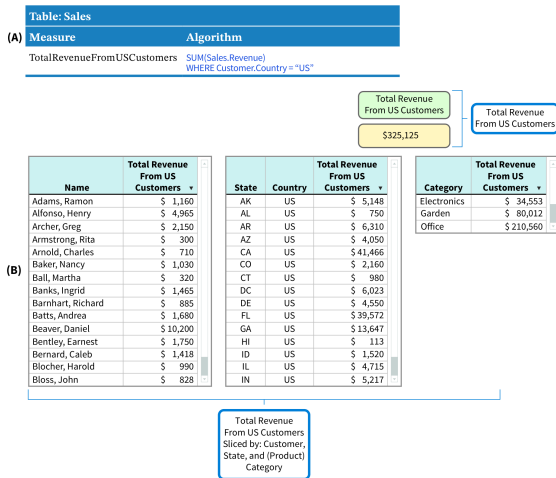
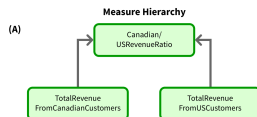


ILLUSTRATION 6.19



Measure Hierarchy to Calculate the Canadian/USRevenueRatio Measure

Table: Sales	
Measures	Algorithm
TotalRevenueFromCanadianCustomers	SUM(Revenue) WHERE Customer.Country = "CA"
TotalRevenueFromUSCustomers	SUM(Revenue) WHERE Customer.Country = "US"
Canadian/USRevenueRatio	TotalRevenueFromCanadianCustomers / TotalRevenueFromUSCustomers

(B)

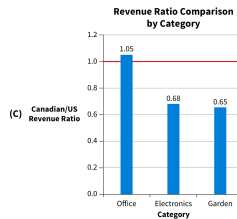


ILLUSTRATION 6.20

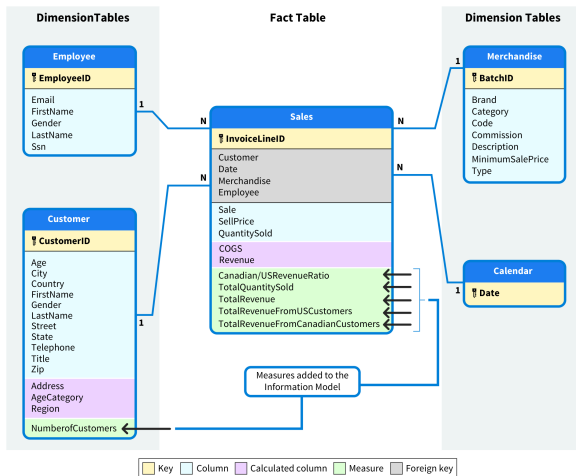


ILLUSTRATION 6.21

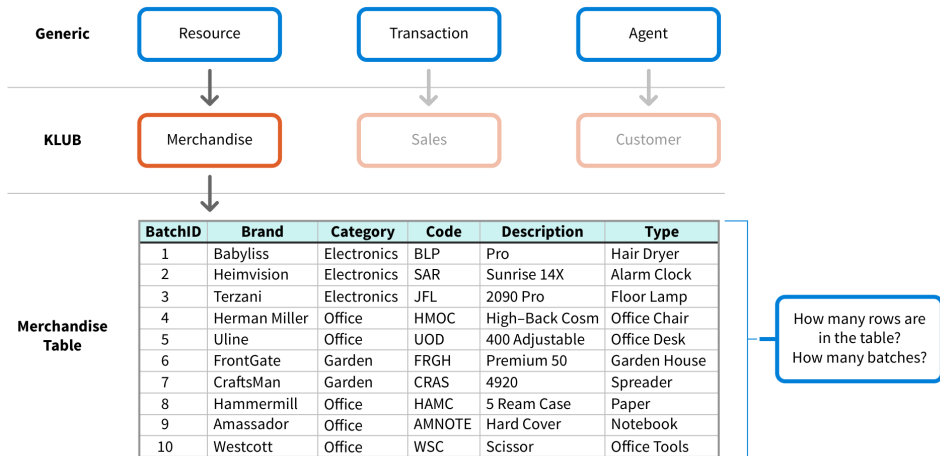
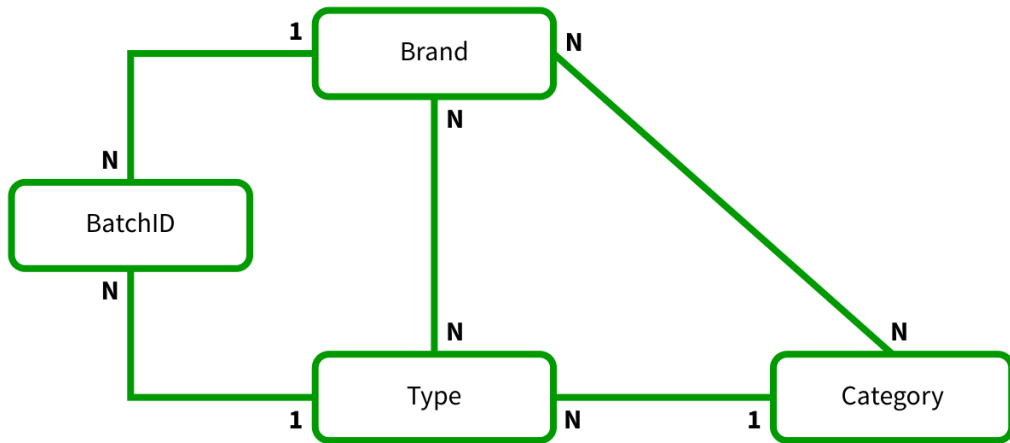


ILLUSTRATION 6.22



6.2 Bảy Mẫu Thuật toán Mô hình hóa Thông tin

- **Mẫu 1 - Tính toán Cơ bản:** Cộng, trừ, nhân, chia (ví dụ: Tổng Doanh thu).
- **Mẫu 2 - Logic và Điều kiện (Logic/Conditional):** Hàm IF, CASE WHEN (ví dụ: Phân loại Nợ xấu nếu quá hạn 90 ngày).
- **Mẫu 3 - Xử lý Văn bản (Text):** Nối chuỗi, tách chuỗi, làm sạch văn bản (ví dụ: Tách Mã Vùng từ Số Điện thoại).

ILLUSTRATION 6.23

Table: Vendor	
Measure	Algorithm
NumberOfVendors	COUNT(VendorID)

Table: Sales	
Measure	Algorithm
NumberOfSales	COUNT(InvoiceLineID)

Table: Purchase	
Measure	Algorithm
NumberOfPurchases	COUNT(BatchID)

Table: Employee	
Measure	Algorithm
NumberOfEmployees	COUNT(EmployeeID)

Table: Merchandise	
Measure	Algorithm
NumberOfBatches	COUNT(BatchID)

ILLUSTRATION 6.24

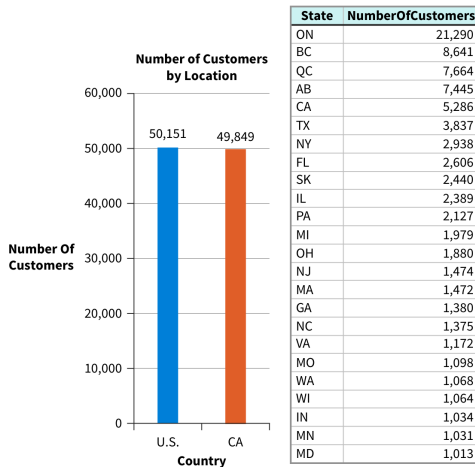


ILLUSTRATION 6.25

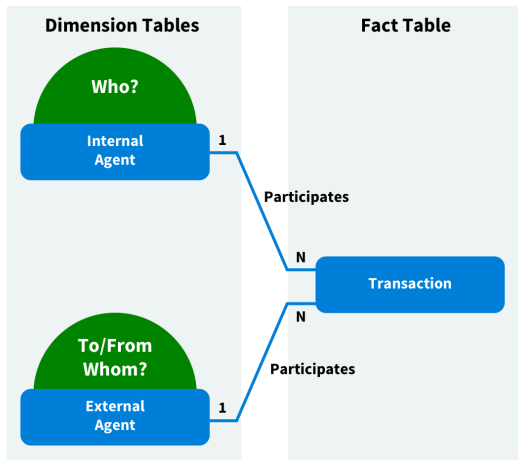


ILLUSTRATION 6.26

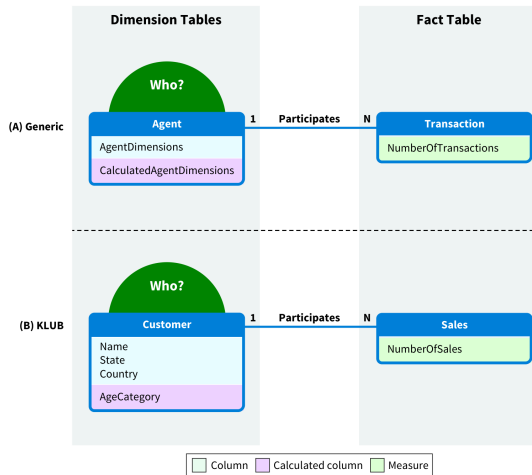


ILLUSTRATION 6.27

1 What is the number of sales transactions per customer (name)?	5 What is the number of sales transactions per age category?
2 Who are the customers (name) with the highest number of sales transactions?	6 What the number of sales per employee (name)?
3 What is the number of sales transactions per state (customer)?	7 Who are the employees (name) with the highest number of sales transactions?
4 What is the number of sales transactions per country (customer)?	8 Who are the customers (name) with a number of transactions of at least two?

Dimensions

Measures

ILLUSTRATION 6.28

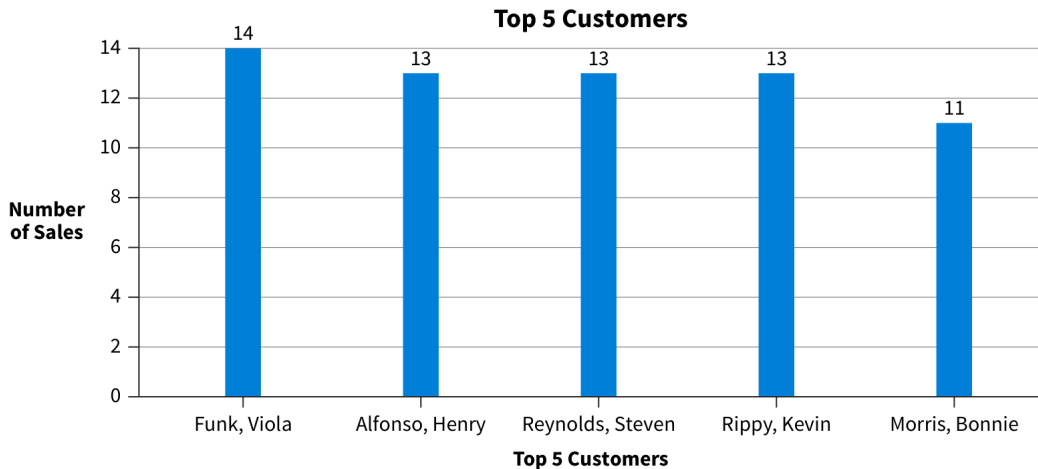


ILLUSTRATION 6.29

Name	NumberOfSales
Funk, Viola	14
Alfonso, Henry	13
Reynolds, Steven	13
Rippy, Kevin	13
Morris, Bonnie	11
Moore, Steven	9
Williams, Mary	9
Zebrowski, Connie	9
Garza, Jeffrey	8
Uren, Harold	8
Barnhart, Richard	7
Edens, Blaine	7
Jenkins, Toni	7
Ward, Jenny	7
Younce, Stacey	7

ILLUSTRATION 6.30

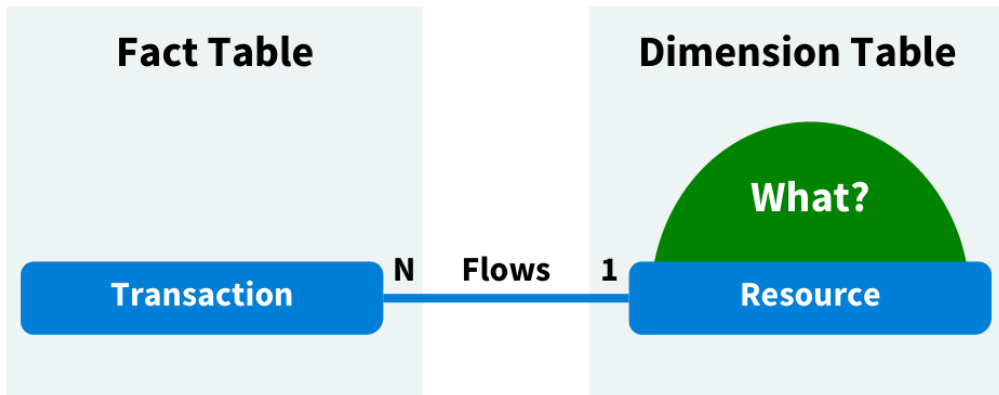


ILLUSTRATION 6.31

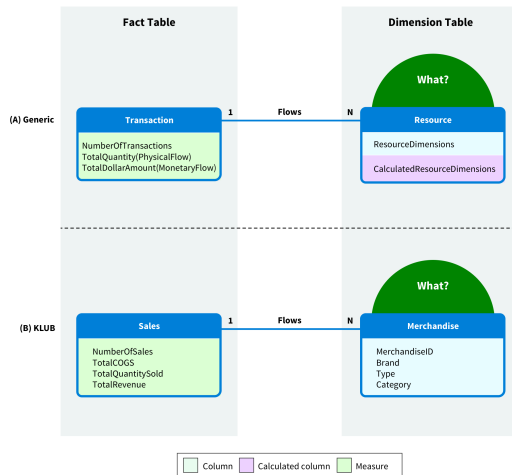


ILLUSTRATION 6.32

Table: Purchases

Measure

Algorithm

TotalQuantityBought

SUM(QuantityBought)

Table: Sales

Measure

Algorithm

TotalQuantityOfPrintersSold

SUM(QuantitySold) WHERE Type = "Printer"

ILLUSTRATION 6.33

- | | |
|--|--|
| 1 What is the number of sales transactions per batch (MerchandiseID)? | 8 What is the total cost per batch (MerchandiseID)? |
| 2 What is the number of units bought per batch (MerchandiseID)? | 9 What is the total revenue generated per batch (MerchandiseID)? |
| 3 What is the number of units sold per batch (MerchandiseID)? | 10 What is the total revenue generated per brand ? |
| 4 What is the number of items sold per brand ? | 11 What is the total revenue generated per type ? |
| 5 What is the number of items sold per type ? | 12 What is the total revenue generated per category ? |
| 6 What is the number of items sold per category ? | 13 What is the total number of printers (type) sold? |
| 7 What is the relative contribution of each category to the total revenue generated? | |

■ Dimensions ■ Measures

Các Mẫu Thuật toán tiếp theo

- **Mẫu 4 - Thời gian (Date/Time):** Trích xuất Tháng, Năm, Quý (ví dụ: Doanh thu theo Quý).
- **Mẫu 5 - Hàm Tài chính (Financial):** Tính PV, FV, NPV, IRR.
- **Mẫu 6 - Tỷ suất (Ratios):** Tính ROA, ROE, Current Ratio.
- **Mẫu 7 - Hàm Phức hợp (Complex/Nested):** Kết hợp nhiều hàm với nhau.

ILLUSTRATION 6.34

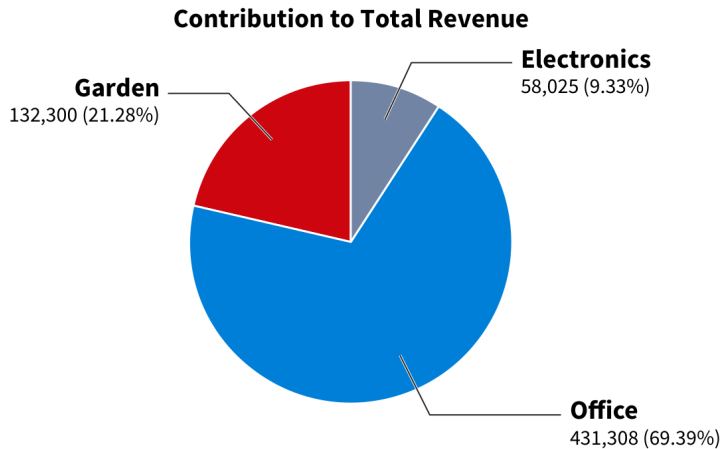


ILLUSTRATION 6.35

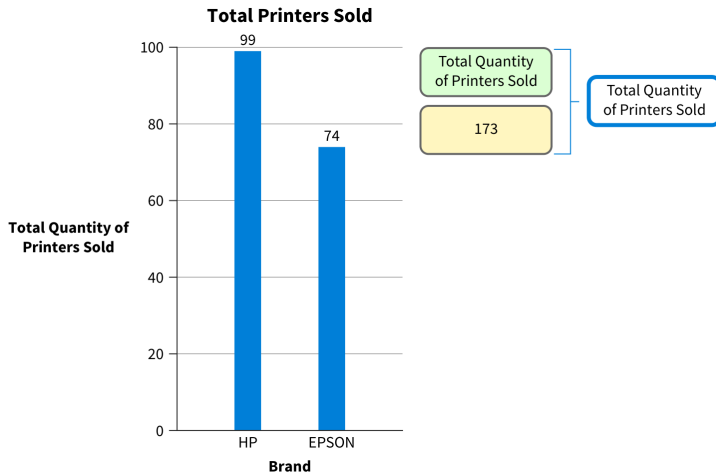


ILLUSTRATION 6.36

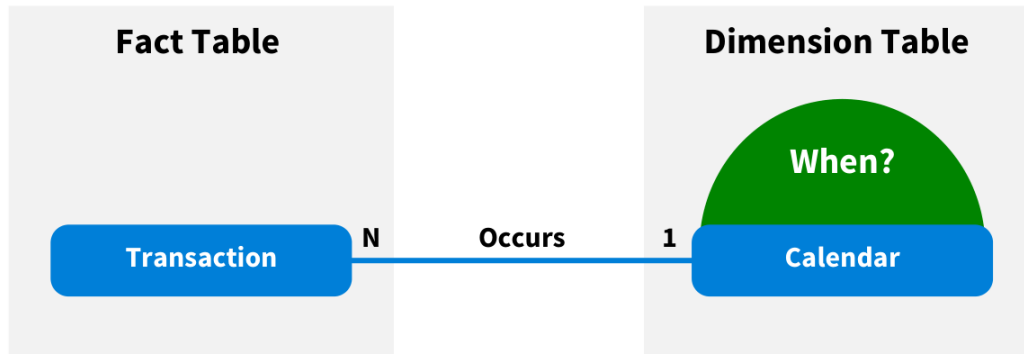


ILLUSTRATION 6.37

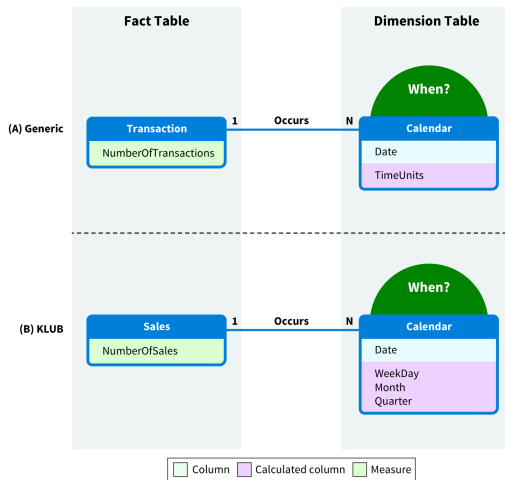


ILLUSTRATION 6.38

1

How do the **number of sales transactions** compare across **weekdays**?

2

How do the **number of purchase transactions** compare across **weekdays**?

3

What is the trend for the **number of sales transactions** during the first six **months** of 2023?

4

How do the **number of sales transactions** for the **first and second quarters** compare?



Dimensions



Measures

ILLUSTRATION 6.39

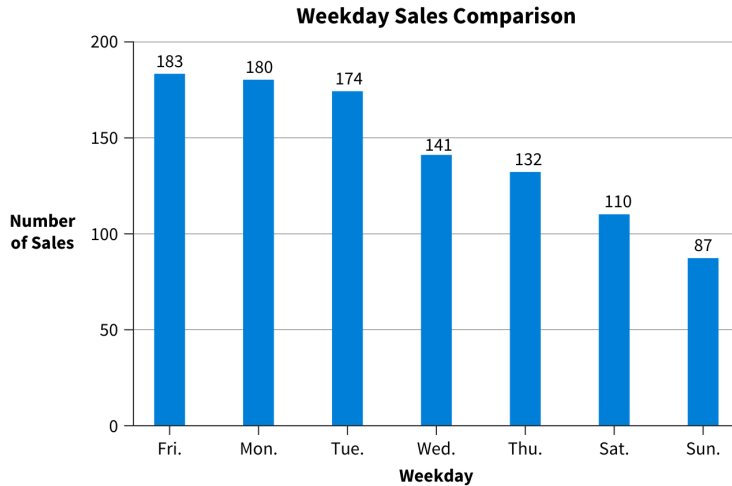


ILLUSTRATION 6.40

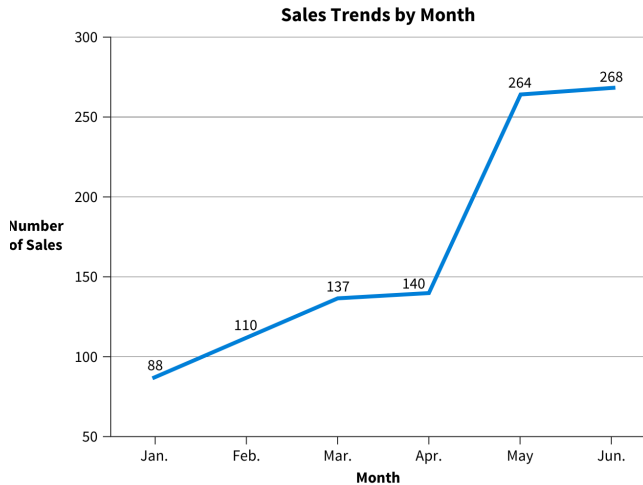


ILLUSTRATION 6.41

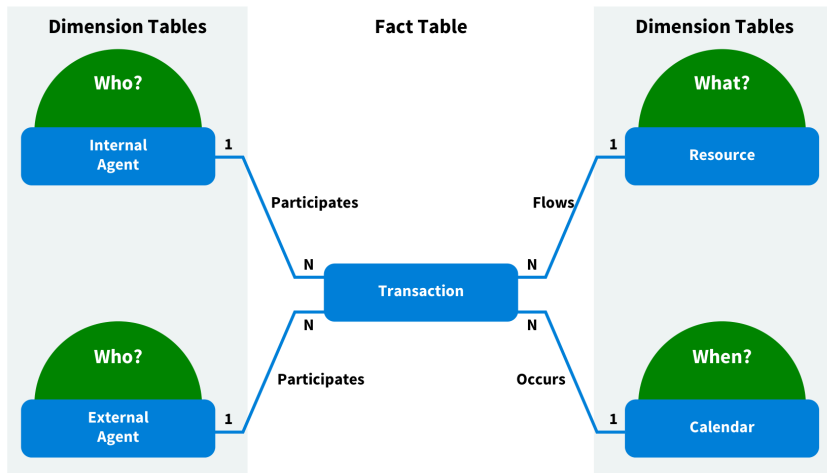


ILLUSTRATION 6.42

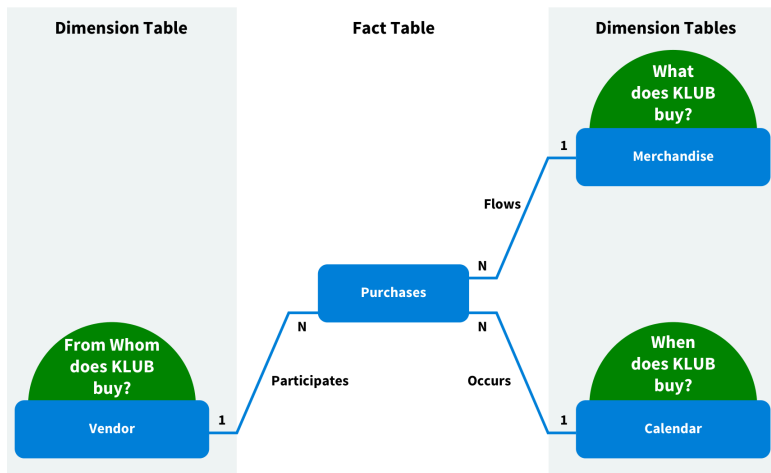


ILLUSTRATION 6.43

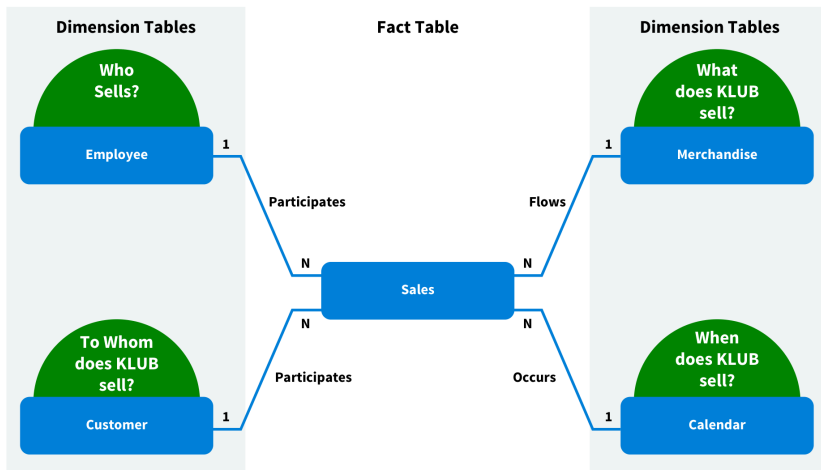
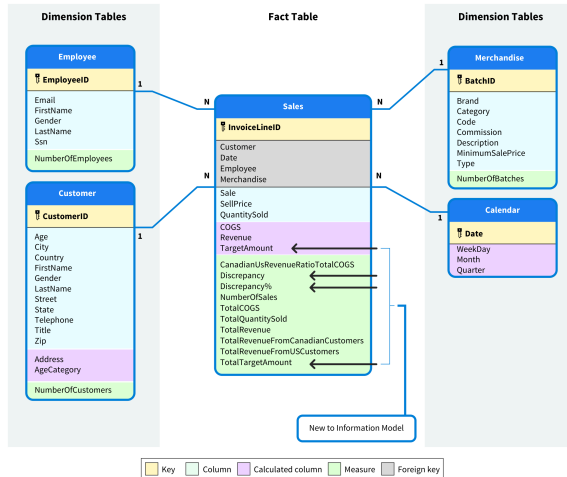


ILLUSTRATION 6.44



6.3 Sáu Mẫu Cấu trúc Dữ liệu Kế toán

- **Mẫu 1 - Hệ thống Tài khoản (Chart of Accounts):** Bảng danh mục cốt lõi của mọi hệ thống kế toán.
- **Mẫu 2 - Dữ liệu Giao dịch (Transactions):** Bảng Fact chứa các bút toán nhật ký (Sổ cái).
- **Mẫu 3 - Danh mục Thực thể (Entity/Master Data):** Khách hàng, Nhà cung cấp, Sản phẩm, Nhân viên.

ILLUSTRATION 6.45

Table: Sales	
Calculated Column	Algorithm
TargetAmount	$\text{Sales.QuantitySold} \times \text{Merchandise.MinimumSalePrice}$

Table: Sales	
Measures	Algorithm
TotalTargetAmount	$\text{SUM}(\text{TargetAmount})$
Discrepancy	$[\text{TotalRevenue}] - [\text{TotalTargetAmount}]$
Discrepancy%	$[\text{TotalRevenue}] / [\text{TotalTargetAmount}]$

ILLUSTRATION 6.46

1

What is the **total quantity sold** per **category** (what) during weekends (**weekday**) (when) to female (**gender**) customers living in the **Mid-Atlantic region** (**state**) (who)?

2

Compare the **total revenue** generated by **brand** (what), **age category** and **gender** (who)?

3

Compare the **discrepancy %** across product **categories** (what)—which category exceeds revenue expectations most?

4

Determine the **discrepancy %** for employees (**name**) (who)—who is the better negotiator?

5

Determine the **discrepancy %** for employees (**name**) (who) across **categories** (what)—who should sell what?



Dimensions



Measures

ILLUSTRATION 6.47

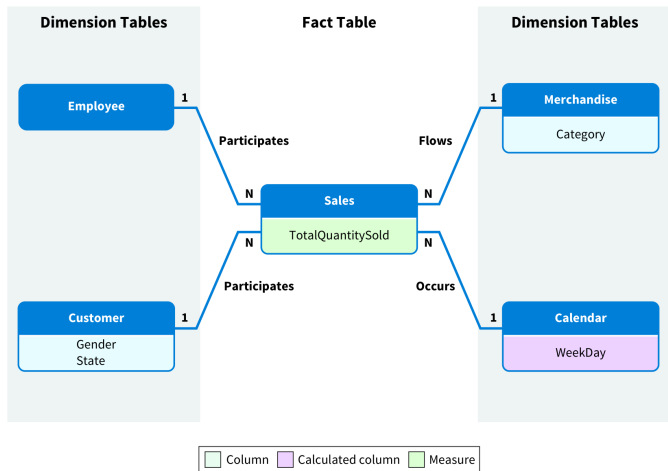


ILLUSTRATION 6.48

Table: Sales	
Measure	Algorithm
QuantitySoldOnWeekdaysToFemaleMidAtlanticCustomers	<pre>SUM(QuantitySold) WHERE Calendar.Weekday = 1 or 7, AND Customer.Gender = "Female", AND Customer.State = "PA" OR "MD" OR "DC" OR "VA" OR "WV" OR "DE" OR "NJ" OR "NY"</pre>

ILLUSTRATION 6.49

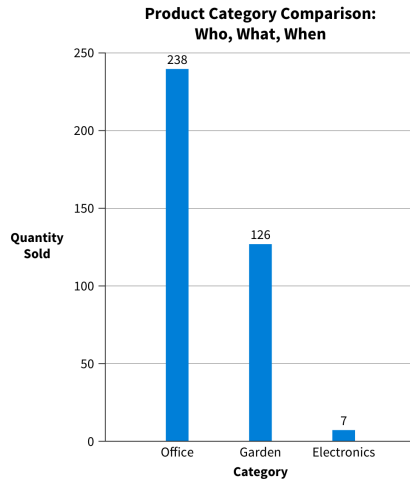


ILLUSTRATION 6.50

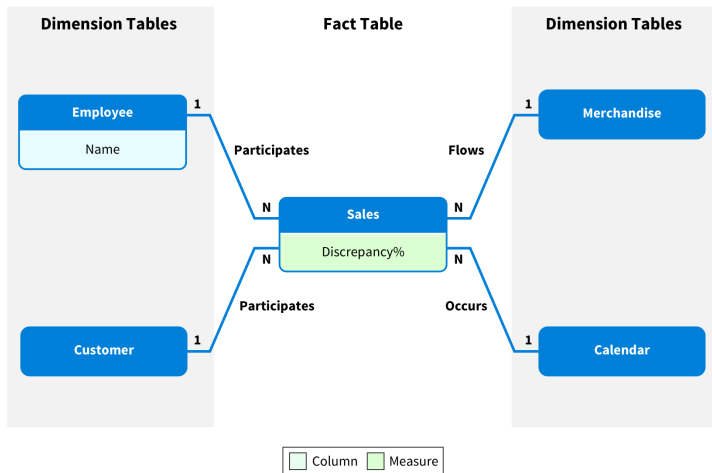


ILLUSTRATION 6.51

EmployeeName	Discrepancy%
Gianni, Kobierski	12.06
Jade, Steeden	4.80
Arlyn, McCreery	3.43
Rosabel, Gabbitus	2.94
Dalenna, Patillo	2.86
Valery, Bayles	2.86
Catherin, Tolliday	2.66
Haze, Coomer	2.59
Florella, McAllan	2.56
Hermina, Jeandillou	2.47
Padriac, Parlour	2.44
Alyda, Greenhall	2.40
Tamra, Lambdin	2.38
Berti, Pridding	2.36
Loreen, Polye	2.36
Catharine, Laimable	2.35
Audie, Braniff	2.33
Charlie, Libri	2.27
Clim, Orhtmann	2.25
Chancey, Donke	2.25
Aarika, Bircher	2.20
Edik, Knowles	2.19
Silvain, MacTague	2.07
Sally, Blizard	1.94
Timmie, Brookzie	1.93

ILLUSTRATION 6.52

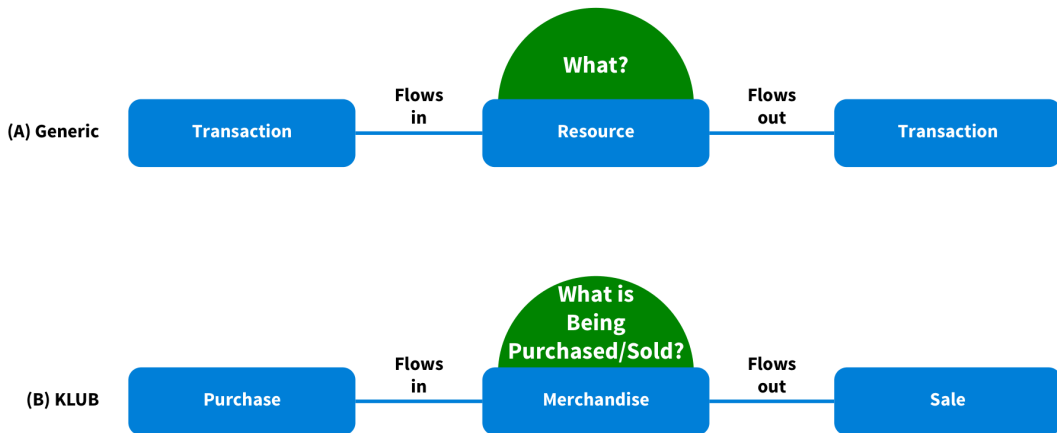


ILLUSTRATION 6.53

Concept/Field	Field Type	Description
Revenue	Calculated Column	The dollar amount paid by the customers for items sold.
COGS	Calculated Column	The dollar amount paid by KLUB for the items they sold.
Profit	Calculated Column	The profit KLUB made on the items they sold.
NetProfit	Calculated Column	The net profit KLUB made on the items sold after paying a commission (additional cost) to their employees (salespeople).
TotalRevenue	Measure	Total revenue generated from sales.
TotalCOGS	Measure	Total dollar amount paid by KLUB for items they sold.
TotalProfit	Measure	Total profit generated from sales.
TotalNetProfit	Measure	Total net profit generated from sales. Commissions for employees are considered.
OverallNetProfit Margin	Measure	The overall net profit margin for KLUB's sales. Total net profit / total revenue.

ILLUSTRATION 6.54

Table: Sales	
Calculated Columns	Algorithm
Profit	Revenue – COGS
NetProfit	$\text{Sales.Profit} - (\text{Sales.Revenue} \times (\text{Merchandise.Commission})/100)$

Table: Sales	
Measure	Algorithm
TotalProfit	$\text{SUM}(\text{Profit})$
TotalNetProfit	$\text{SUM}(\text{NetProfit})$
OverallNetProfitMargin	$[\text{TotalNetProfit}] / [\text{TotalRevenue}]$

ILLUSTRATION 6.55

Table: Sales

Measure

Algorithm

ProfitsForUSElectronicsSales

TotalProfit Where Customer.Country = “US” AND
Merchandise.Category = “Electronics”

Cấu trúc Dữ liệu (Tiếp theo)

- **Mẫu 4 - Ngân sách so với Thực tế (Budget vs Actual):** Kết nối bảng kế hoạch và bảng thực tế để phân tích chênh lệch (Variance Analysis).
- **Mẫu 5 - Bảng Thời gian (Time Dimension):** Cực kỳ quan trọng để phân tích xu hướng (Time Intelligence) trong Power BI / Tableau.
- **Mẫu 6 - Sổ cái và Sổ chi tiết (General & Subsidiary Ledgers):** Mối quan hệ 1-N (One-to-Many) giữa Sổ cái tổng hợp và các Sổ chi tiết (Phải thu, Phải trả, Hàng tồn kho).

ILLUSTRATION 6.56

1

What is the **overall net profit margin** for the different product **categories**?

2

Who are the 25 **customers** (name) with the **highest net profit margin**?

3

Which printer **brand**, HP or Epson, has the highest **profit margin**?

4

Which employees (**name**) have generated at least \$1,000 in **profit**?

5

How do Canadian provinces (**state**) rank based on **total profit** generated?

6

What is the trend (**month**) in **profits** generated by selling electronics (**category**) to U.S. (**state**) customers?



Dimensions



Measures

ILLUSTRATION 6.57

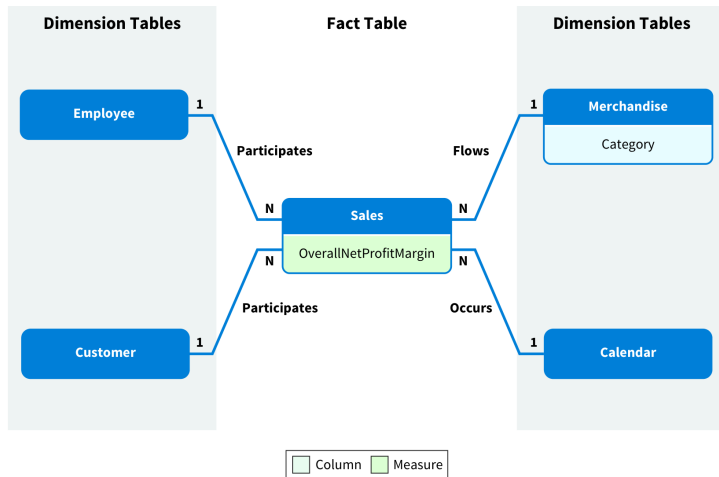


ILLUSTRATION 6.58

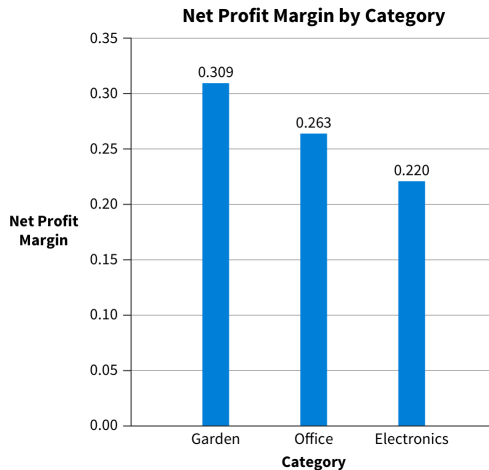


ILLUSTRATION 6.59

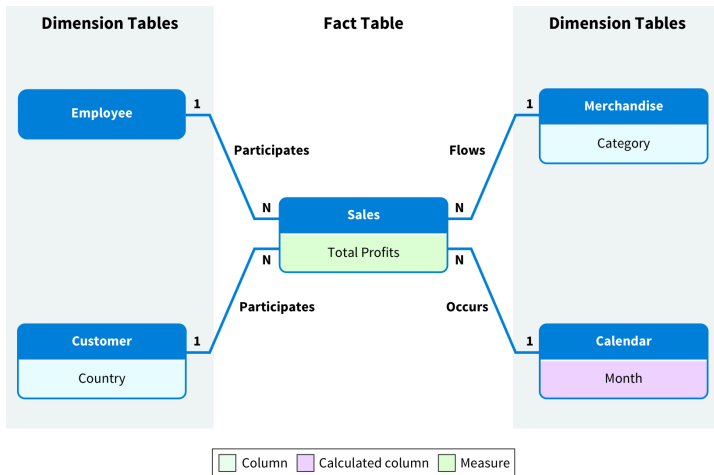


ILLUSTRATION 6.60

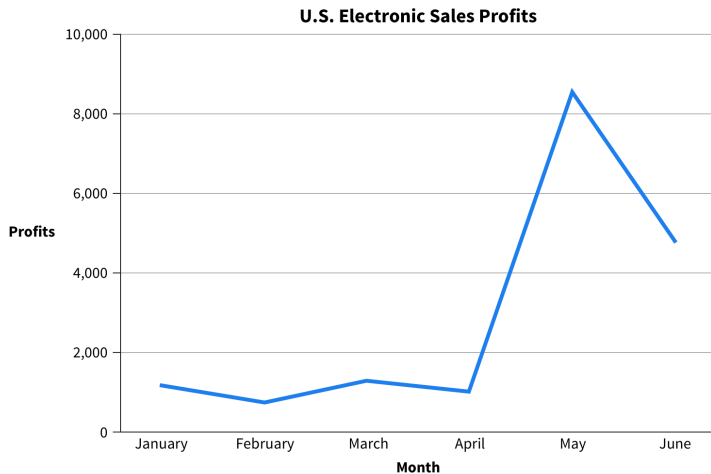


ILLUSTRATION 6.61A

ILLUSTRATION 6.61 Overview of Information Modeling Patterns

Id	Algorithm	Description	Implementation
Implementation Patterns			
1	Within-Table Numeric Calculation	Creates a new calculated column by applying arithmetic operators to columns in the same table.	Basic arithmetic operators: addition, subtraction, multiplication, division, or exponentiation.
2	Within-Table Text Calculation	Creates a new calculated column by applying text manipulation operators to columns in the same table.	Text manipulation operators such as concatenate, extract, or replace.
3	Within-Table Classification	Creates a new calculated column that contains new knowledge by applying conditional logic.	Boolean expressions and Boolean operators such as AND, OR, and NOT.
4	Across-Table Calculation	Data from different, linked tables are used to create a new calculated column.	Navigation operators such as joins.
5	Single-Column Aggregation	Creates a new measure that aggregates all the values of a single column.	Aggregate functions such as SUM, COUNT, AVERAGE, MIN, and MAX.
6	Filtered Aggregation	Creates a new measure that aggregates a subset of the values of a single column.	Apply filters. Examples are SUMIF, COUNTIF, and AVERAGEIF functions in Excel or the WHERE clause in SQL.
7	Measure Hierarchies	Creates a new, more complex measure using existing measures.	Apply arithmetic operators to measures such as addition, subtraction, multiplication, or division.

(Continued)

ILLUSTRATION 6.61B

ID	Information	Description	Implementation
Content Patterns			
8	How Many	Analyzes how many of a concept exists.	Create measures that counts rows in tables.
9	Participates Transaction-Who	Analyzes who is involved in a transaction.	Create measures for transactions and calculated columns for agents.
10	Flows Transaction-What	Analyzes what is involved in a transaction.	Create measures for transactions and calculated columns for resources.
11	Occurs Transaction-When	Analyzes when a transaction occurred.	Create measures for transactions and calculated columns for calendars.
12	Who-What-When Star Schemas	Integrates analysis of the who, what, and when transactions.	Create measures for transactions and calculated columns for agents, resources, and calendars.
13	Integrated Star Schemas	Analyze how inflows and outflows affect resources.	Create measures for transactions and calculated columns for resources.

ILLUSTRATION 6.62

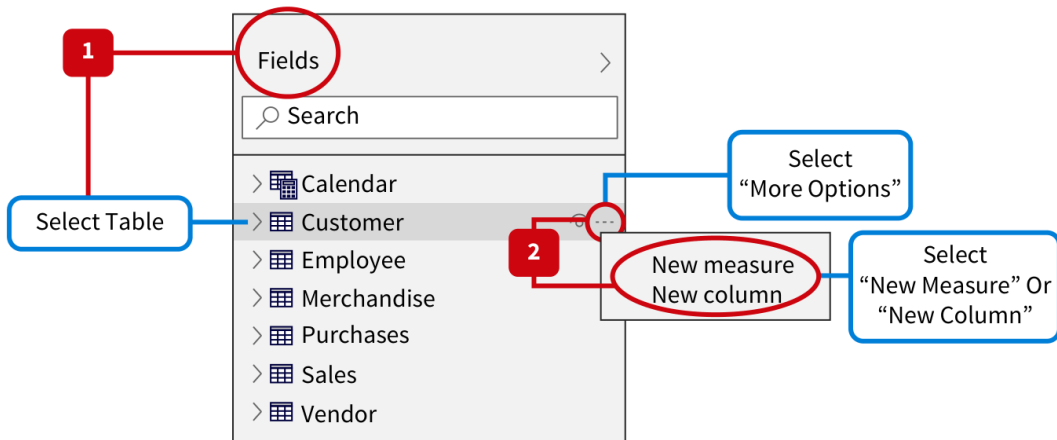


ILLUSTRATION 6.63

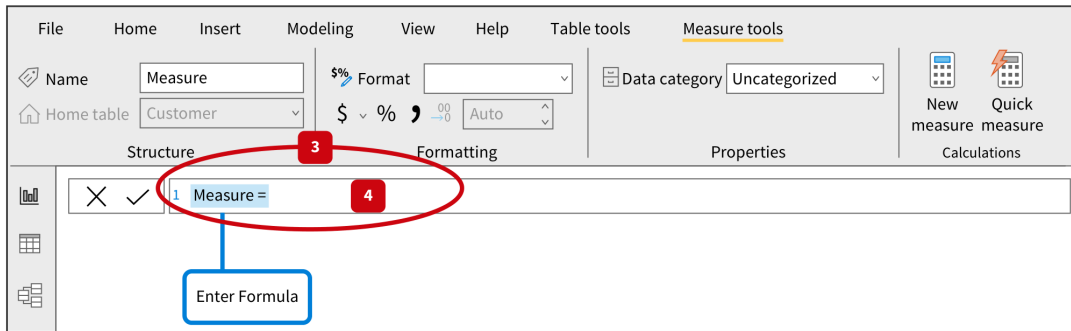


ILLUSTRATION 6.64

Query: Total Revenue from US Customers by Product Category

```
SELECT Category, SUM([SellPrice] * [QuantitySold]) AS [TotalrevenueFromUSCustomers]
FROM Merchandise INNER JOIN (Customer INNER JOIN Sales ON Customer.CustomerID = Sales.Customer)
ON Merchandise.BatchID = Sales.Merchandise
WHERE Country="US"
GROUP BY Category
```

Detailed Descriptions Code

Code	Description
<code>SUM([SellPrice] * [QuantitySold])</code>	Calculates the total revenue generated from sales.
<code>WHERE Country="US"</code>	A filter defined through a Boolean expression. Only sales from U.S. customers are considered.
<code>GROUP BY Category</code>	The total revenue is broken down by category. There are three values for CATEGORY: Electronics, Garden, and Office.
<code>FROM Merchandise INNER JOIN (Customer INNER JOIN Sales ON Customer.CustomerID = Sales.Customer) ON Merchandise.BatchID = Sales.Merchandise</code>	Defines the joins between the tables and thus the navigation path.

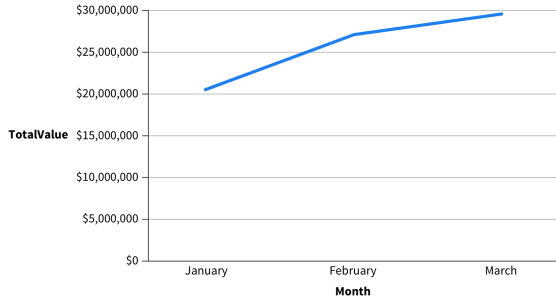
ILLUSTRATION 6.65

	Category ▼	Total Revenue From US Customers ▼
	Electronics	34,553
	Garden	80,012
	Office	210,560

Phần Bài tập Ngắn

Brief Exercises (BE)

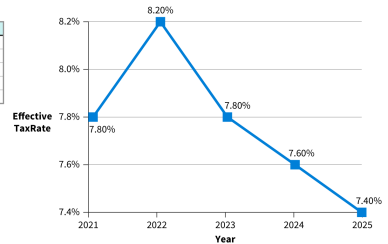
BE 6.5 & BE 6.11



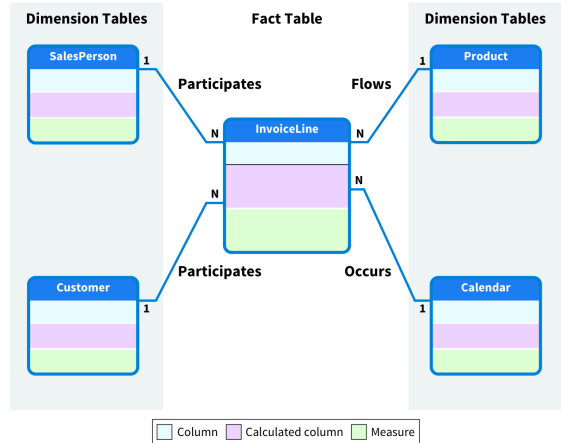
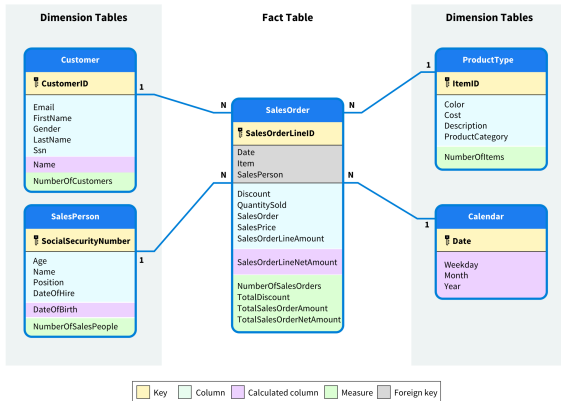
Year	Account	Amount
2021	Income Before Tax	\$ 14,365,450
2021	Income Tax Expense	\$ 1,120,505
2022	Income Before Tax	\$ 15,299,804
2022	Income Tax Expense	\$ 1,254,584
2023	Income Before Tax	\$ 165,998,110
2023	Income Tax Expense	\$ 12,947,853
2024	Income Before Tax	\$ 13,489,990
2024	Income Tax Expense	\$ 1,025,239
2025	Income Before Tax	\$ 175,009,898
2025	Income Tax Expense	\$ 12,950,732

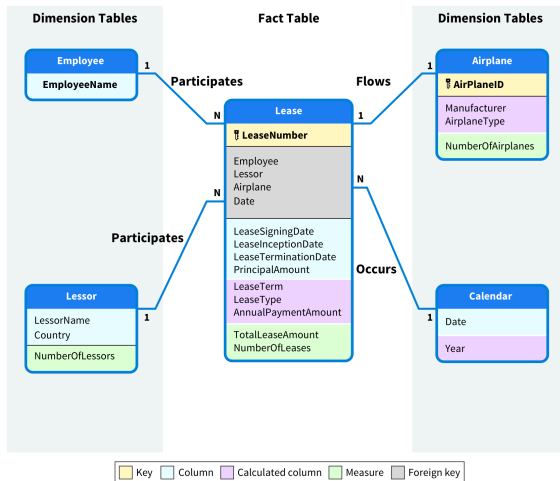
1. Develop the information model that calculates the effective tax rate.
2. Use it to create the following report.

Year	EffectiveTaxRate
2021	7.80%
2022	8.20%
2023	7.80%
2024	7.60%
2025	7.40%



BE 6.12 & BE 6.14





Phần Bài tập

Exercises (EX)

EX 6.2 & EX 6.3

Total Net Revenue

\$3,319,960

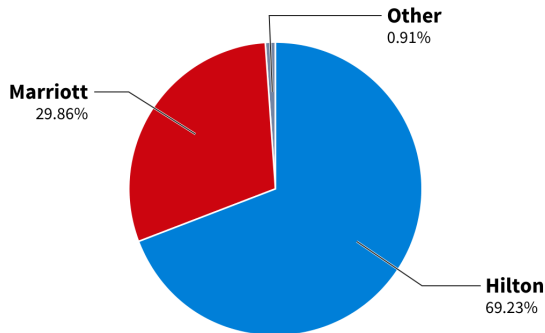
Total Tax

\$136,463.17

FinishedGood	RawMaterial	STDQuantity	STDCost
Tricycle	Trike tire, large front	1	\$ 96
Tricycle	Tricycle frame	1	\$ 150
Tricycle	4" tire	1	\$ 64
Tricycle	Tricycle pedals	2	\$ 28
Mountain bike	18" Frame	1	\$ 213
Mountain bike	26" wheels and tires	2	\$ 34
Mountain bike	Mountain Bike handle	2	\$ 44
Mountain bike	Gear system	1	\$ 81
Buggy	Nuts	2,500	\$ 0.01
Buggy	Bolts	2,750	\$ 0.01
Buggy	Radio	1	\$ 111
Buggy	Truck body	1	\$ 430
Buggy	Engine	1	\$ 1,015
Buggy	Automotive tires	4	\$ 69
Buggy	Black Paint	40	\$ 2

EX 6.4A & EX 6.4B

FinishedGood	TotalStandardCost
Buggy	\$ 1,964.50
Mountain bike	\$ 450.00
Tricycle	\$ 366.00



EX 6.5 & EX 6.5_1

Hotel	TotalAmountSpent
Hilton	\$ 170,154.52
Marriott	\$ 73,376.80
Other	\$ 2,225.38
Total	\$ 245,756.70

Candidate	Age	Audit	Tax	Level
Liz Smith	33	Y	N	4
Jada Vaughn	29	Y	Y	3
Jenny Wang	52	N	N	1
Eric Kim	47	Y	Y	3
Misty Harrison	41	N	Y	4
George Star	39	Y	N	5
Elton Seger	25	N	N	3
Isabella Fernande	38	Y	N	2
Elise Jackson	41	Y	Y	2
Brenda Swift	25	N	N	5
Lindsay Lauper	56	N	Y	4
Nicole Sardo	35	N	N	2
Cindy Rafferty	40	N	N	2
Greg Cuddy	45	Y	N	3
Mateo Rodriguez	31	N	Y	4
Caleb Cooke	28	Y	N	2
Jane Harris	41	N	Y	5
Debra Pollack	29	Y	N	1
Sheldon Andrews	49	Y	Y	2

EX 6.7A & EX 6.7B

Name	2021	2022	2023	2024	2025
ABC Groceries	7.55%	7.94%	7.55%	7.14%	8.05%
Neighborhood Groceries	6.83%	6.76%	7.02%	6.67%	6.78%
Worldwide Groceries	7.62%	8.57%	7.52%	8.27%	8.59%

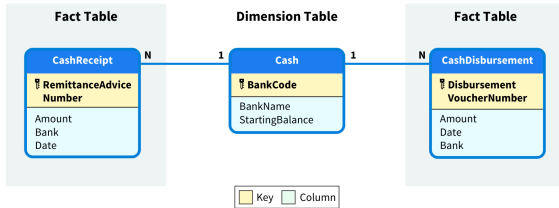
Name	Profit Margin 2021-2025
ABC Groceries	7.65%
Neighborhood Groceries	6.82%
Worldwide Groceries	8.08%

EX 6.9 & EX 6.10

TruckID	MaximumWeight	TotalWeight	Load
1	10,000	7,826.40	0.78
2	25,000	25,974.83	1.04
3	15,000	11,794.72	0.79
4	20,000	16,997.62	0.85
5	20,000	8,267.33	0.41
6	25,000	19,246.33	0.77
7	15,000	4,078.55	0.27

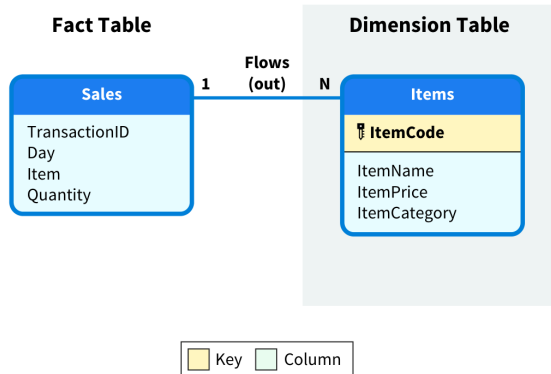
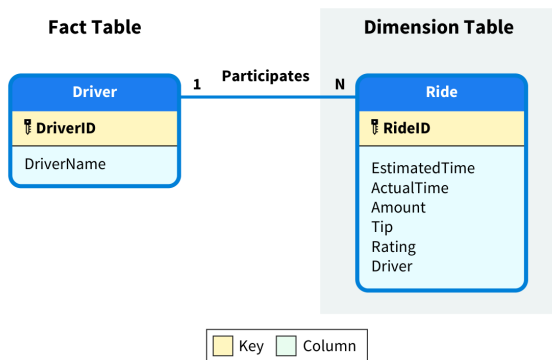
Employee	Total Vested Contribution
David Maurer	705,047.00
Mike Cook	638,946.00
Bernard Espinosa	438,663.75
Susan Hoover	437,302.25
Jamal Coleman	321,413.75
Elzbieta Jacek	234,108.00
Gail Wolcott	78,167.25
Billy Houston	50,912.40
Carlos Martinez	37,590.80
Amy Winter	30,483.70

EX 6.11A & EX 6.11B



BankName	StartingBalance	Inflow	Outflow	CustomerBalance
Fulton	\$ 23,009	\$ 953.34	\$ 1,148.36	\$ 22,813.98
MTB	\$ 89,012	\$ 2,116.00	\$ 900.23	\$ 90,227.77
PNC	\$ 44,099	\$ 2,408.04	\$ 2,987.36	\$ 43,519.68
Total	\$ 156,120	\$ 5,477.38	\$ 5,035.95	\$ 156,561.43

EX 6.12 & EX 6.13A



Sales

Name	Description
TransactionID	An ID used to identify the transaction.
Day	The day of the week the transaction occurred.
Item	The item sold to the customer (code).
Quantity	The quantity sold of the item.

Items

Name	Description
ItemCode	A code that uniquely identifies a type of product.
ItemName	The name of the item (product type).
ItemPrice	The item's price. All items have a fixed price.
ItemCategory	The item's category.

Tình huống Ứng dụng Chuyên môn

Professional Application Cases (PAC)

PAC 6.1 & PAC 6.4

Category	Hourly Rate
Apprentice	\$ 45
Intermediate	\$ 65
Advanced	\$ 85
Champion	\$ 110

		Month	Deduction
Salary	200,000	January	1,275.00
FICA%	0.0765	February	1,275.00
SocialSecurity%	0.062	March	1,275.00
Medicare%	0.0145	April	1,275.00
SocialSecurityMaximum	168,900	May	1,275.00
		June	1,275.00
		July	1,275.00
		August	1,275.00
		September	1,275.00
		October	1,275.00
		November	380.13
		December	241.67

LO 1.4

TransactionID	Date	Amount	Loyalty	LoyaltyDiscount
1	2/1/2025	\$ 144.17	Y	1.44
2	2/1/2025	\$ 35.80	N	0
3	2/2/2025	\$ 289.09	N	2.89
4	2/2/2025	\$ 134.88	Y	1.35
5	2/3/2025	\$ 99.50	N	1

Scenario	Facts	Dimensions		
		What	Who	When
a.	Total revenue	Product category	Customer	Month
b.	Total revenue		Region	
c.	Total profit	Product type	Region	Weekday
d.	Total number of customers		Region	
e.	Total number of sales	Product category	Customer	
f.	Total number of sales			Month
g.	Total cost	Product type		

Algorithm

IF ((Age < 30 AND LoyaltyProgram = "Yes") OR (Sales > 10,000 AND NumberOfIssues = 0))
THEN "A"
ELSE "B"

Data set

CustomerID	Name	Age	LoyaltyProgram	Sales	NumberOfIssues
1	Kinsun	32	Yes	\$ 13,833	0
2	Barbara	29	No	\$ 15,000	2
3	Jon	29	Yes	\$ 15,000	2
4	Gabriella	25	Yes	\$ 20,000	0

a. 1

b. 2

c. 3

d. 4

Algorithm

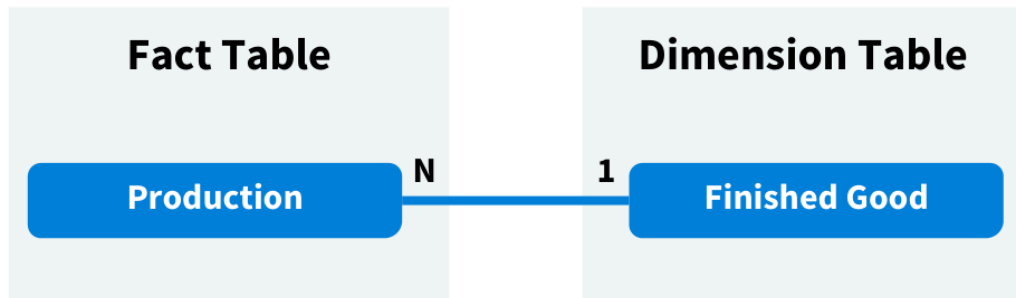
PRICE TYPE =

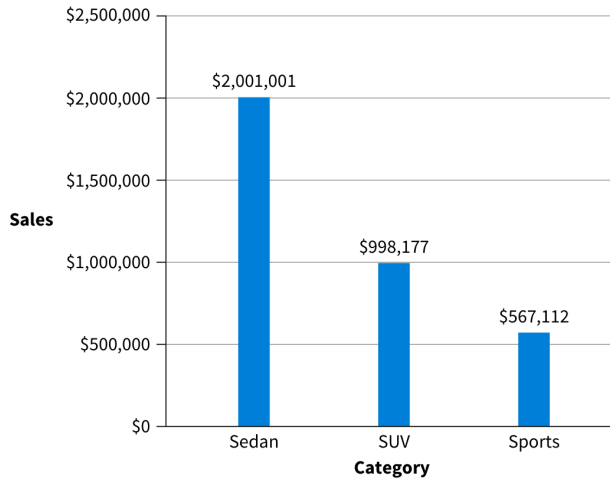
IF Price < 25,000 THEN “LOW-END” ELSE

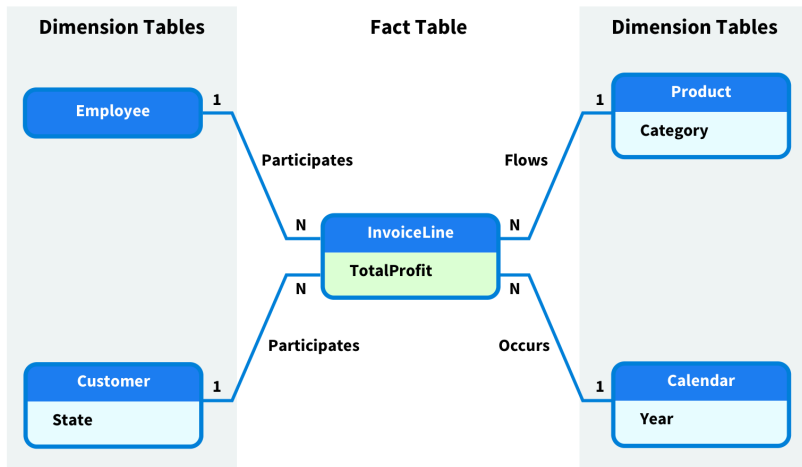
IF Price >= 25,000 AND PRICE < 50,000 THEN “MID-RANGE” ELSE
“HIGH-END”

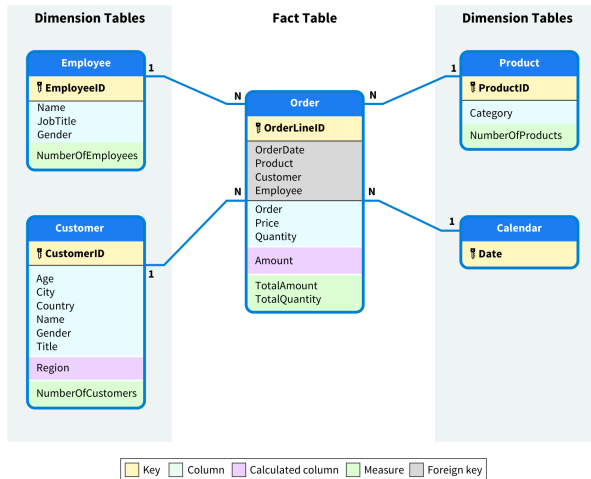
Data set

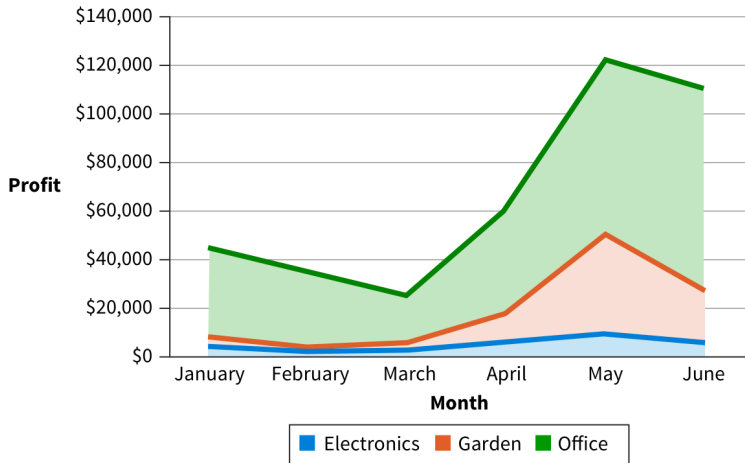
ProductID	Description	Year	Price
1	VW Beetle	2019	\$ 14,322
2	Honda Pilot	2018	\$ 25,119
3	BMW X7	2019	\$ 50,432
4	Chevrolet Cru..	2019	\$ 11,654
5	Mercedes ML	2017	\$ 29,750
6	Ford Explorer	2019	\$ 32,881
7	Honda Civic	2017	\$ 9,811
8	Chevrolet Tah..	2016	\$ 42,889
9	Jeep Wrangler	2007	\$ 9,999
10	Maserati Leva..	2020	\$ 69,922

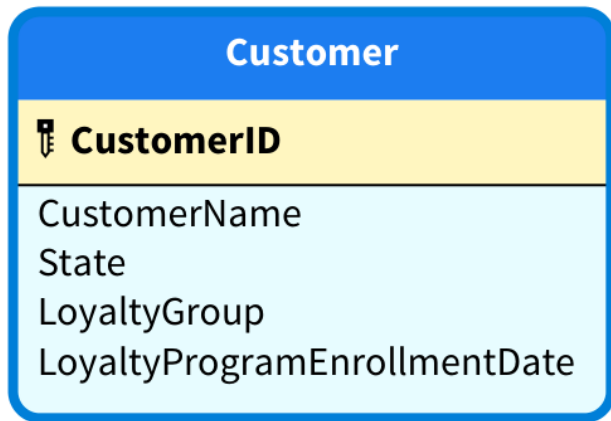












DTunes data dictionary 6.0

Instructors

Column Name	Field Description
InstructorID	An ID that uniquely identifies an instructor.
InstructorName	An instructor's name.
NumberOfHoursTaught	The number of hours an instructor has taught for D*Tunes.
NumberOfAwards	The number of local and regional awards an instructor has received during the last three years.
NumberOfNationalAwards	The number of national recognitions an instructor has received.
Salary	For full-time employees, the salary the instructor earns.
JobTitle	An instructor's job title.

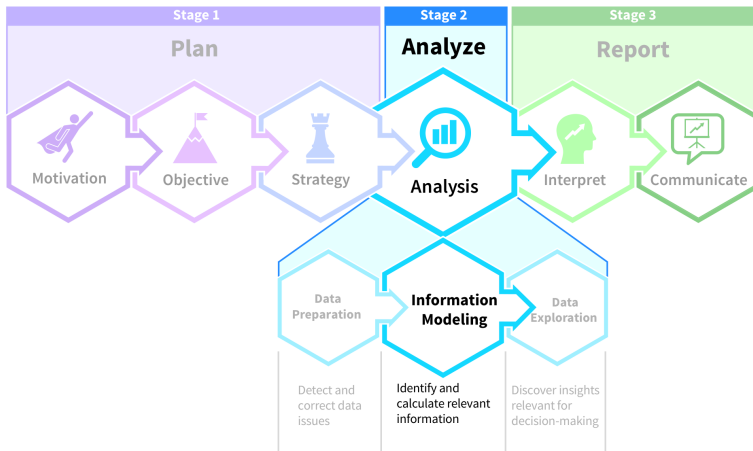
Classes

Column Name	Field Description
ClassCode	A code that uniquely identifies a class.
ClassDescription	A brief description of the class.

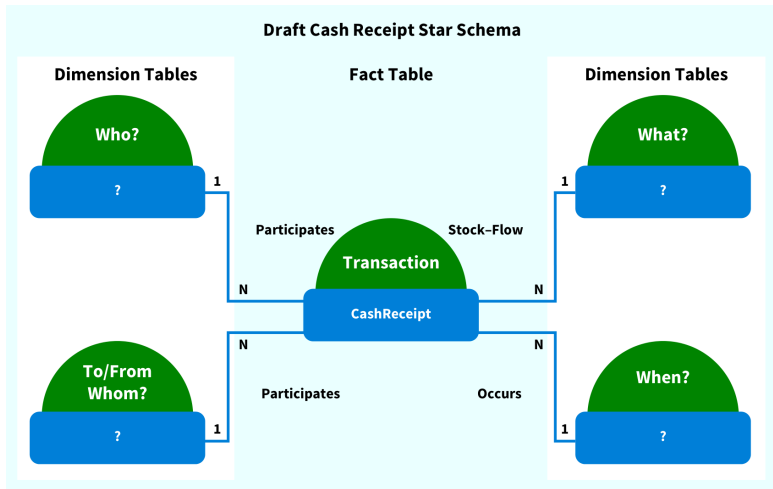
Students

Column Name	Field Description
StudentID	An ID that uniquely identifies a student.
StudentName	A student's name.
Level	Student's Skill Level: B = Beginner I = Intermediate A = Advanced
Gender	A student's gender.

Fig 6.0



Apply It 6.1



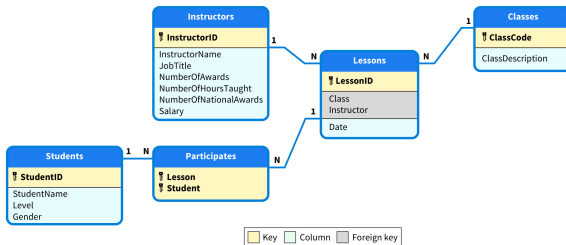
DTunes data dictionary 6.1

Lessons

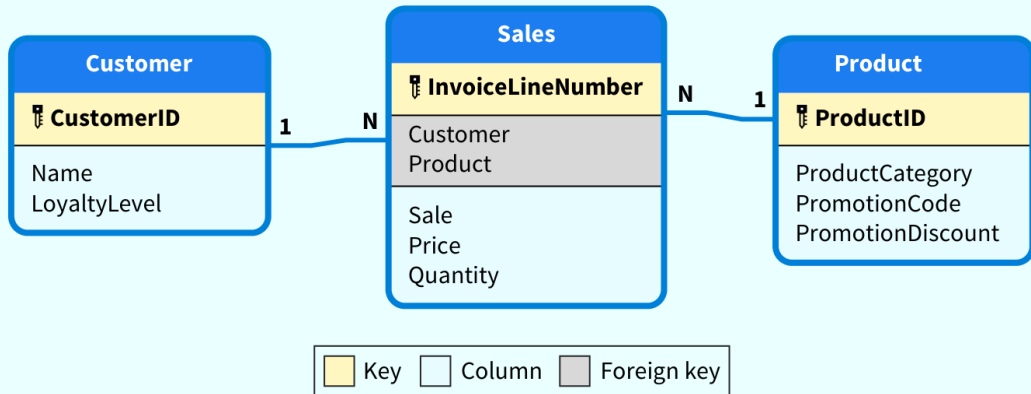
Column Name	Field Description
LessonID	An ID that uniquely identifies a lesson.
Date	The date on which the lesson occurred.
Class	The type of class taught during this lesson.
Instructor	The ID of the instructor teaching this class.

Participates

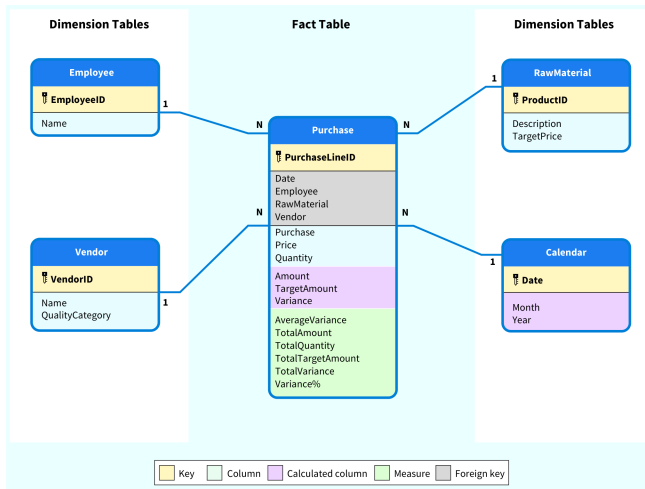
Column Name	Field Description
Lesson	The ID of the lesson taken by the student.
Student	The ID of the student taking the lesson.



Apply It 6.2



Apply It 6.3



Hỏi & Đáp

Cảm ơn các bạn đã lắng nghe!